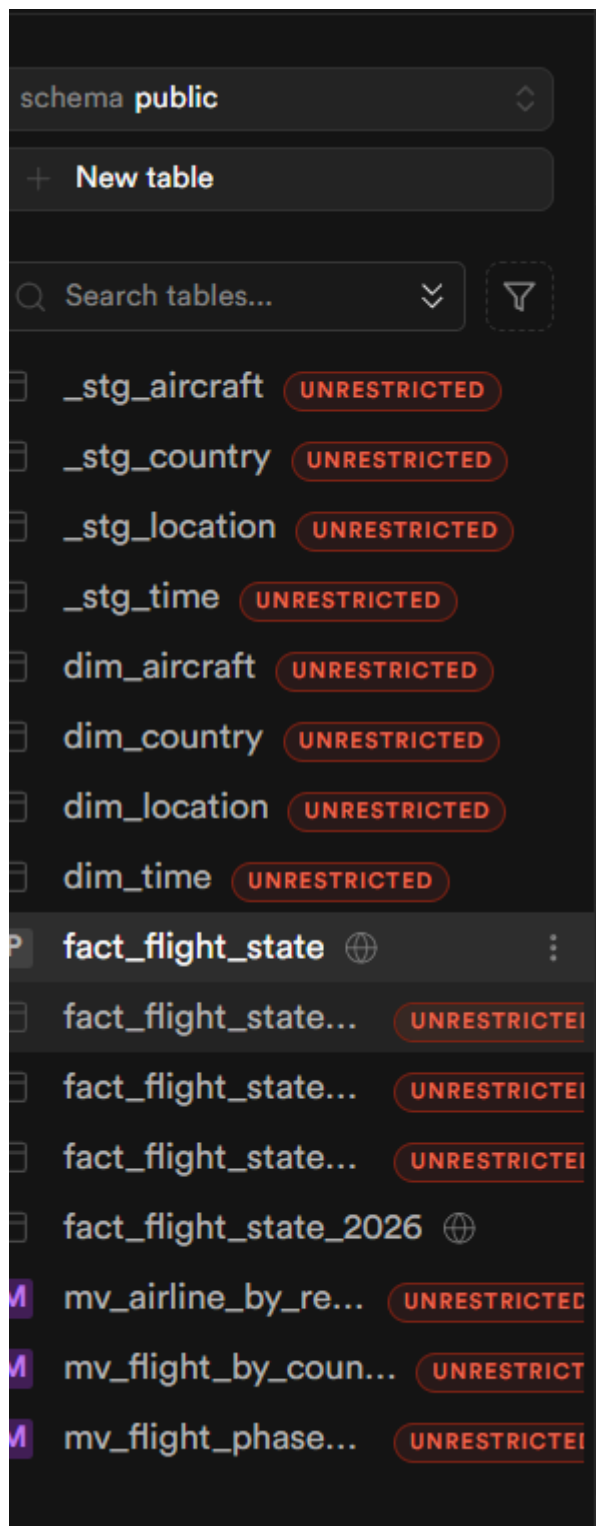
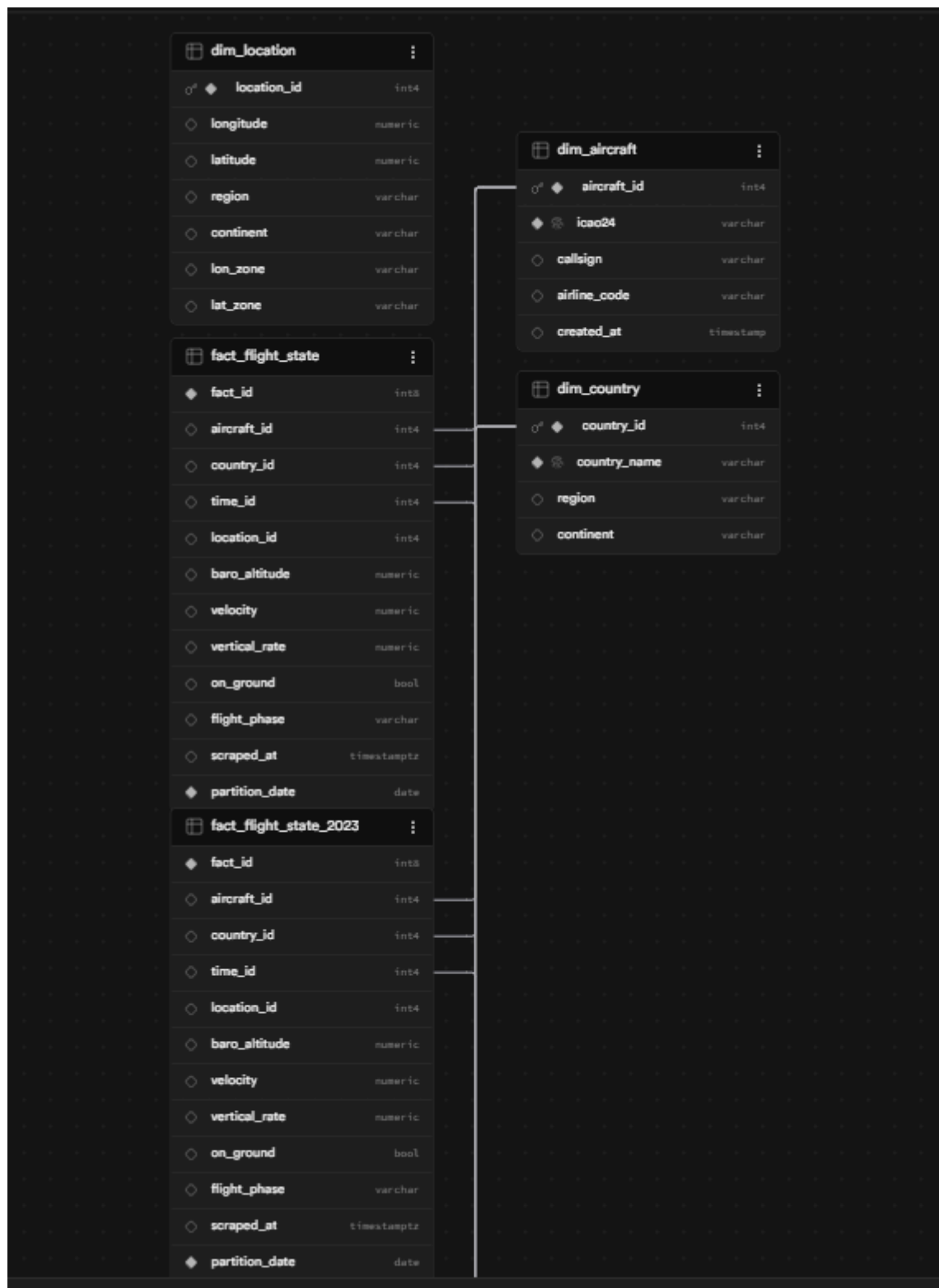
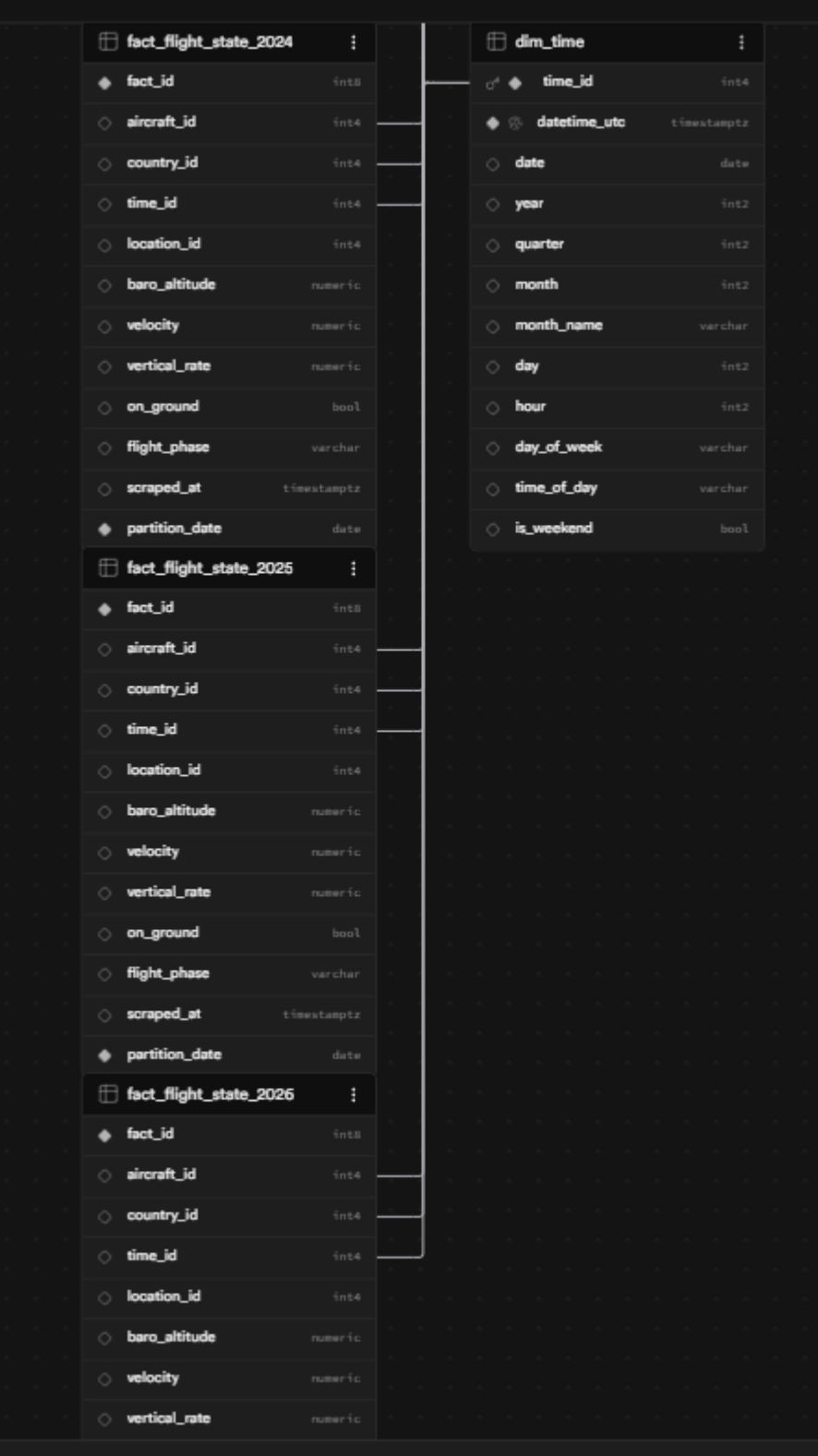


ISI database:



Schema keseluruhan:





 _stg_aircraft	⋮
◇ icao24	text
◇ callsign	text
◇ airline_code	text

 _stg_country	⋮
◇ country_name	text
◇ region	text
◇ continent	text

 _stg_time	⋮
◇ datetime_utc	timestampz
◇ date	date
◇ year	int4
◇ quarter	int4
◇ month	int4
◇ month_name	text
◇ day	int4
◇ hour	int4
◇ day_of_week	text
◇ time_of_day	text
◇ is_weekend	bool

 _stg_location	⋮
◇ longitude	float8
◇ latitude	float8
◇ region	text
◇ continent	text
◇ lon_zone	text
◇ lat_zone	text

Simulasi Periodik:

```
import os

N_BATCHES          = 4384
INTERVAL_HOURS     = 6
START_DATE         = datetime(2023, 1, 1, 0, 0, 0)
ROWS_PER_BATCH     = 150
ADD_NOISE          = True
RANDOM_SEED         = 42
```

Extension Postgre:

```
DDL_STATEMENTS = [  
    # Extensions  
    'CREATE EXTENSION IF NOT EXISTS pg_stat_statements',  
    'CREATE EXTENSION IF NOT EXISTS btree_gist',  
  
    # dim_aircraft  
    '''CREATE TABLE IF NOT EXISTS dim_aircraft (  
        aircraft_id SERIAL PRIMARY KEY,  
        icao24          VARCHAR(10) NOT NULL UNIQUE,  
        callsign       VARCHAR(20),  
        airline_code   VARCHAR(5),  
        created_at     TIMESTAMP DEFAULT NOW()  
    )''',  
  
    # dim_country  
    '''CREATE TABLE IF NOT EXISTS dim_country (  
        country_id SERIAL PRIMARY KEY,  
        country_name VARCHAR(100) NOT NULL UNIQUE,  
        region        VARCHAR(50),  
        continent     VARCHAR(50)  
    )''',  
  
    # dim_time  
    '''CREATE TABLE IF NOT EXISTS dim_time (  
        time_id SERIAL PRIMARY KEY,  
        datetime_utc TIMESTAMP WITH TIME ZONE NOT NULL UNIQUE,  
        date      DATE,  
        year      SMALLINT,  
        quarter   SMALLINT,  
        month     SMALLINT,  
        month_name VARCHAR(10),  
        day       SMALLINT,  
        hour      SMALLINT,  
        day_of_week VARCHAR(10),  
        time_of_day VARCHAR(10),  
        is_weekend BOOLEAN
```

```

# dim_location
'''CREATE TABLE IF NOT EXISTS dim_location (
    location_id SERIAL PRIMARY KEY,
    longitude    NUMERIC(9,5),
    latitude     NUMERIC(9,5),
    region       VARCHAR(50),
    continent    VARCHAR(50),
    lon_zone     VARCHAR(10),
    lat_zone     VARCHAR(10),
    CONSTRAINT dim_location_lon_lat_unique UNIQUE (longitude, latitude)
)''',

# fact_flight_state - location_id sekarang punya REFERENCES
'''CREATE TABLE IF NOT EXISTS fact_flight_state (
    fact_id      BIGSERIAL,
    aircraft_id  INT REFERENCES dim_aircraft(aircraft_id),
    country_id   INT REFERENCES dim_country(country_id),
    time_id      INT REFERENCES dim_time(time_id),
    location_id  INT REFERENCES dim_location(location_id),
    baro_altitude NUMERIC(10,2),
    velocity     NUMERIC(8,2),
    vertical_rate NUMERIC(8,2),
    on_ground    BOOLEAN,
    flight_phase VARCHAR(15),
    scraped_at   TIMESTAMP WITH TIME ZONE,
    partition_date DATE NOT NULL
) PARTITION BY RANGE (partition_date)''',

```

```

# Partisi per tahun
'''CREATE TABLE IF NOT EXISTS fact_flight_state_2023
    PARTITION OF fact_flight_state
    FOR VALUES FROM ('\2023-01-01\') TO ('\2024-01-01\')''',
'''CREATE TABLE IF NOT EXISTS fact_flight_state_2024
    PARTITION OF fact_flight_state
    FOR VALUES FROM ('\2024-01-01\') TO ('\2025-01-01\')''',
'''CREATE TABLE IF NOT EXISTS fact_flight_state_2025
    PARTITION OF fact_flight_state
    FOR VALUES FROM ('\2025-01-01\') TO ('\2026-01-01\')''',
'''CREATE TABLE IF NOT EXISTS fact_flight_state_2026
    PARTITION OF fact_flight_state
    FOR VALUES FROM ('\2026-01-01\') TO ('\2027-01-01\')''',

# BRIN Index
'CREATE INDEX IF NOT EXISTS idx_fact_brin_date    ON fact_flight_state USING BRIN (partition_date)',
'CREATE INDEX IF NOT EXISTS idx_fact_brin_scraped ON fact_flight_state USING BRIN (scraped_at)',

# BTree Index untuk semua FK
'CREATE INDEX IF NOT EXISTS idx_fact_aircraft ON fact_flight_state (aircraft_id)',
'CREATE INDEX IF NOT EXISTS idx_fact_country  ON fact_flight_state (country_id)',
'CREATE INDEX IF NOT EXISTS idx_fact_time     ON fact_flight_state (time_id)',
'CREATE INDEX IF NOT EXISTS idx_fact_location ON fact_flight_state (location_id)',
'CREATE INDEX IF NOT EXISTS idx_fact_phase    ON fact_flight_state (flight_phase)',

```